

JSGP · SHINE HACKATHON

Morning Triage

The morning triage for One Care caseworkers:
who needs you first, and why, in plain language.

THE PROBLEM

Help that must be **asked for**

Alarm buttons and pendants only help while the fallen person can still press them. An unconscious senior presses nothing.

The fall becomes a **long lie**: hours helpless on the floor before anyone thinks to check.

THE APPROACH

Help that needs **no button**

Motion and door sensors log every kitchen visit and every door event. When a morning breaks that pattern, or hours pass with no movement at all, the resident surfaces on the caseload. Nothing worn, nothing pressed, and no camera in the room.

One optional bangle adds a second layer:
falls caught in seconds instead of hours.

All of it is off-the-shelf hardware, and Singapore already deploys senior alert wearables at 26,800-resident scale.

THE TWO DATASETS

Why these two recordings

4,505

REAL RECORDINGS · 1,798 FALLS · 38 PEOPLE · SISFALL

220 days

ONE SENIOR, LIVING ALONE · CASAS ARUBA

One teaches the detector what a fall is in physics.

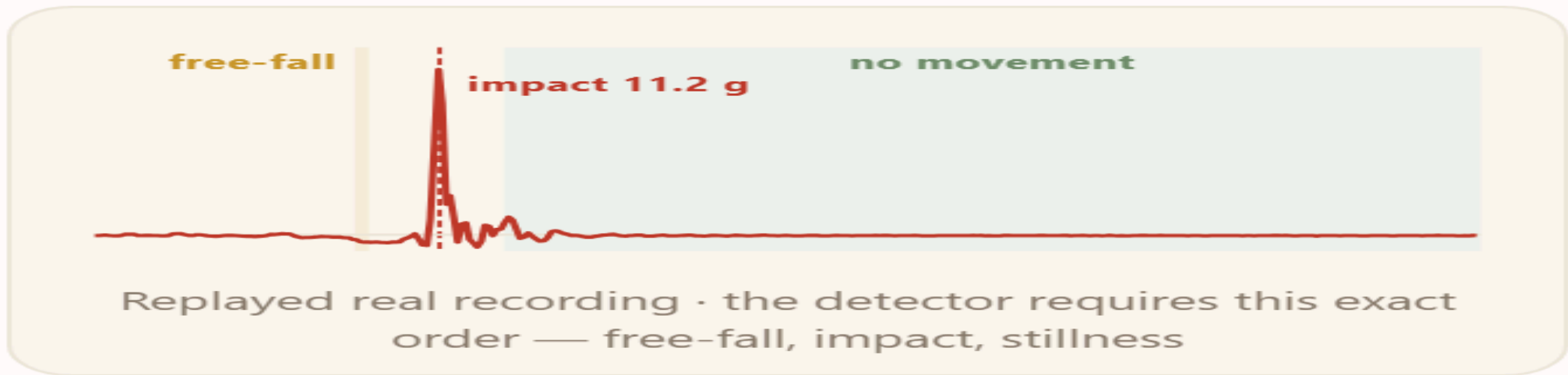
The other teaches the system what one person's normal life is in rhythm.

Both run on the same cheap sensor class a worn bangle carries, the **ADXL345** accelerometer, so what the detector sees in the data is what it would see in a home. Both are checksum-pinned in a committed lock file, and every number reproduces from one script.

HOW IT WORKS

A fall has a **signature**

WHAT THE SENSOR SAW



Straight from the product: the drill-down shows the signal behind every incident

The detector requires this **ordered sequence**: a free-fall dip, an impact spike within half a second, then stillness. A dropped phone cannot fake the order.

WHAT THE NUMBERS SAY

We measured it on **all 4,505**

96.2%

OF 1,798 REAL FALLS DETECTED · SISFALL

18.8%

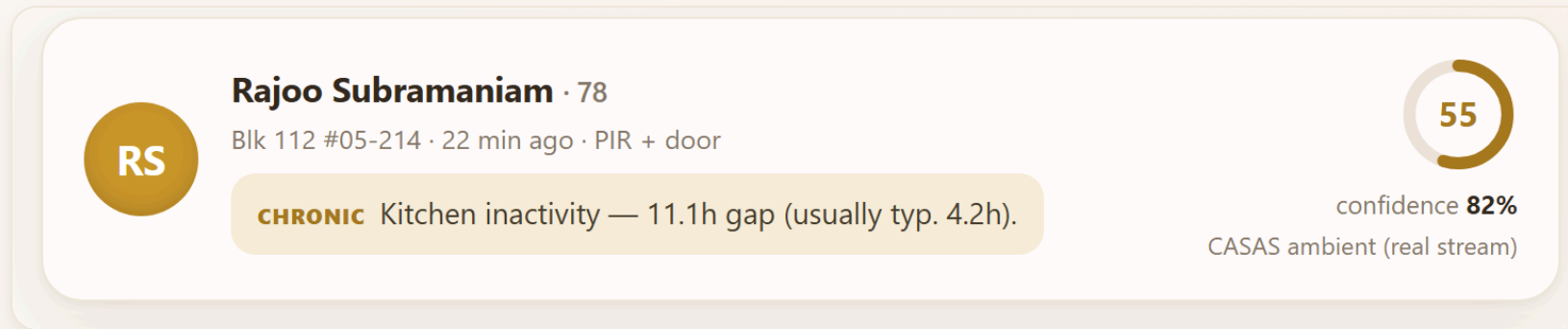
FALSE ALARMS ON ELDERLY MOVEMENT · HALF THE YOUNG-ADULT RATE

We swept **144** threshold combinations over all 4,505 recordings and kept the best trade-off. Nothing was filtered out, and nothing was trained: the numbers come from counting an experiment.

Elderly falls came in at 84.0%, all 75 from a single volunteer, so we hold that figure lightly.

THE OTHER HALF OF THE CASELOAD

220 days taught us **his normal**



Straight from the product: rank 1 carries the 220 real days. The system found the broken day on its own. The **routine track** ranks the calm caseload. From 220 days of one real home it learned that resident's own rhythm: when the kitchen first wakes, how many night trips, when the door first opens. On **2011-02-19**, a date inside that recording, the rhythm broke, and the system pushed him to **rank 1 (0.65)** on its own. We never marked that day. No training, no population average: his history is the baseline.

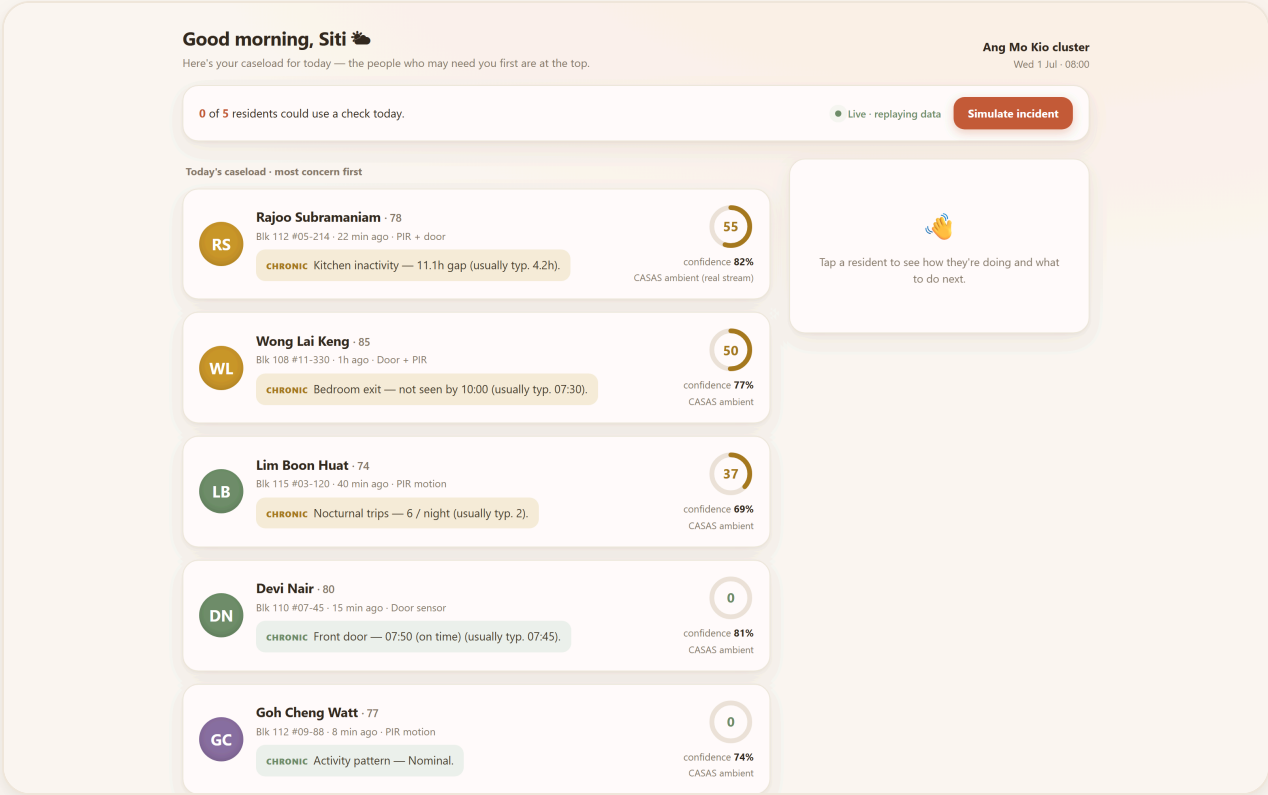
We trained a model **anyway**

The gap is the finding. The model tops out at **80.0%**; the shipped detector reads the same physics at **96.2%**. The 16.2-point gap is the evidence that the temporal order, free-fall then impact then stillness, is what magnitudes alone cannot see.

- We fit a logistic regression on **500** real SisFall recordings, four magnitude features each. **250 falls and 250 ordinary activities on purpose**: an even table keeps accuracy readable at face value, and the committed table re-runs the experiment offline on any machine.
- On the held-out **125** recordings: 80.0% accuracy, 77.9% precision, 84.1% recall. Re-splits move it only 78.5 to 80.0%, and growing the training slice from 38 to 375 recordings moves it only 78.4 to 80.0%: **the size is not the ceiling, the features are.**
- One fit takes **0.050 seconds**, so the `/training` page replays the real recorded run live.

THE PRODUCT

One screen, calm by default



Every resident is ranked by need, and every row explains itself in one sentence.

EXPLAINABLE BY CONTRACT

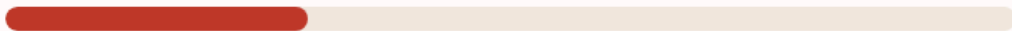
Every number **decomposes**

WHY SHE'S RANKED HERE

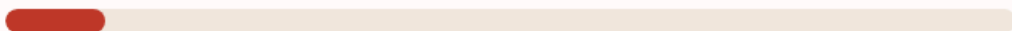
Peak impact **11.2 g**
walking < 1.4 g



Post-impact stillness **12 s**



Orientation **Horizontal**



Every score breaks into the sensor facts that produced it. Nothing is taken on faith

100
ACUTE RISK

100%
CONFIDENCE

Accelerometer
SOURCE

A stale sensor lowers confidence. It never inflates risk, so a dead sensor cannot fake a crisis

The system ranks and explains.

A human decides.

THEN WE ADDED A SECOND ACUTE SOURCE

The camera knows a fall from a **lie-down**

- **A camera catches the fall the bangle misses:** left on the nightstand, or never worn. Pose estimation runs entirely in the browser on `/watch`; upright, then horizontal within **3 s**, then **3 s** of stillness fires the same alert as the bangle. A slower transition is a deliberate lie-down and never alarms.
- Camera incidents carry the label **Camera (pose)** and the heuristic's own 0.55 to 0.85 confidence band. The 96.2% figure belongs to the accelerometer track alone.
- Enrolled face identity is opt-in and on-device, so the caseload row and the alert are **named** and say where to look: last motion: Bedroom. Unmatched people fire the generic alert; no video leaves the browser.

THEN WE CLOSED THE LOOP

The alert **answers back**

- A detected fall sends a **named** Telegram alert: who fell, the unit, and the last-motion area to check first.
- **No recovery within 45 s sends STILL DOWN.** Standing up cancels it, and it fires once per fall.
- A caregiver taps **I am responding**: the alert is stamped with who took it, the first tap wins, and the dashboard shows the name within 5 seconds.
- With no bot configured the alert leg quietly does nothing. Detection never waits on it.

THE DEMO BEAT

A fall preempts everything

0 of 5 residents could use a check today.

● Live · replaying data

Simulate incident

Each press streams a different real recorded fall through the live detector, the same signal a worn bangle would send. A second button injects a dropped phone the detector ignores

FALL DETECTED

TA

Tan Ah Moi · 82
Blk 108 #04-210 · just now · Accelerometer

ACUTE Fall detected — 11.2 g impact then 12 s no movement.

confidence **100%**
Wearable bangle

Call now

Escalate

WHAT TO DO NEXT

CALL NOW. If no response, escalate to MOH 24/7 line + dispatch.

The fall pins to the top in seconds and states its evidence in plain words

The system suggests the next step. The caseworker makes the call

The camera and the alert loop you just saw run live too: `/watch` and the Telegram ping are the same incident path. `/training` replays the trained-model run.

HONEST LIMITS

What we do **not** claim

- It shrinks time to detection. It does not promise to catch every event.
- The senior false-alarm rate is **measured** rather than guessed, because it came in at 18.8% on real elderly recordings, lower than young adults. What stays thin is elderly fall data, since our falls come from one volunteer, and a lab fall is not the same as a home fall.
- A missed fall (or a bangle left on the nightstand) degrades to the routine track, where hours of stillness still surface that resident by morning.
- An irregular life earns wide baselines. The system says so through low confidence instead of hiding it.

THE NEXT STEP

What a pilot would **prove**

- **Escalation beyond the app.** The STILL DOWN follow-up and the I-am-responding acknowledgement are built and live. A pilot adds what sits past the phone: automated welfare calls, and the handoff to the MOH/AIC line when nobody acknowledges.
- **A morning brief per flagged resident.** Auto-drafted wording, grounded strictly in the deterministic features. The drafting layer never touches a score.
- **A single One Care cluster to start.** Measured on two numbers: time to detection, and caseworker minutes saved per shift.

THE ASK

Pilot **with us**

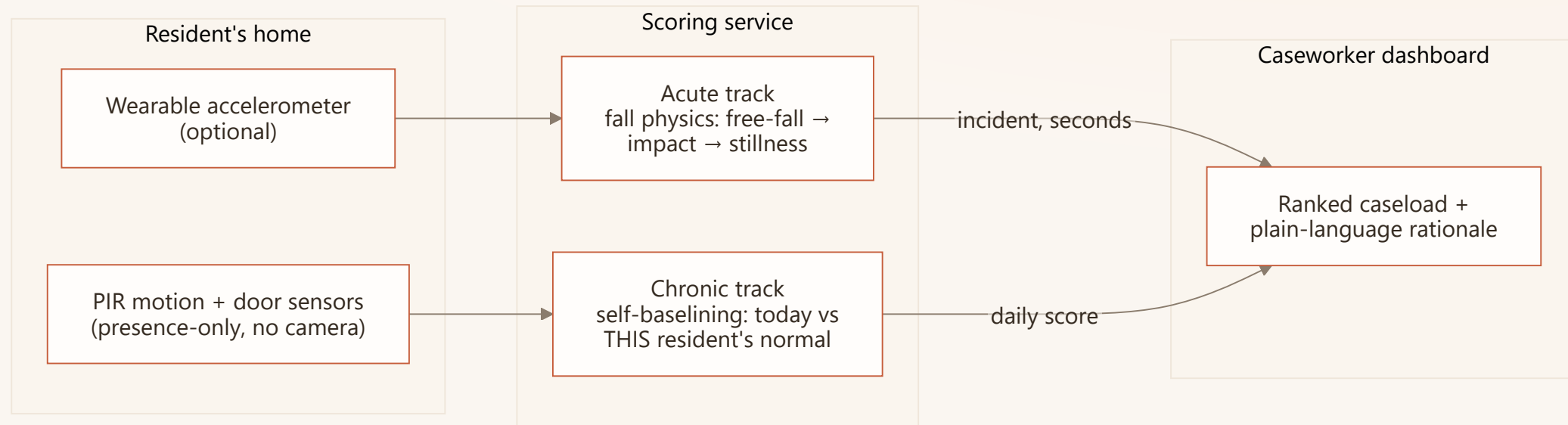
We ask for one cluster and commodity sensors. The working system is already validated on **4,505 real recordings** and a **220-day real home**.

Built at SHINE Hackathon. The demo runs live on real recordings, a different one each press.

Why no machine learning

- **No Singapore training data exists.** A model trained on Colombian or American recordings would import exactly the prior we refuse to assume.
- **A fall is physics.** The free-fall, impact and stillness sequence transfers across bodies and homes without any training step.
- **A caseworker must be able to ask why.** Decomposable rules survive that question. A black box does not.
- Where a model helps later: smoothing the wording of the morning brief. It will never produce a score.

The three layers



The home senses. The service scores. The caseworker decides.

The math is Python, so the service computes and the browser only displays. The API call carries the JSON between them.

From a fall to the **dashboard**

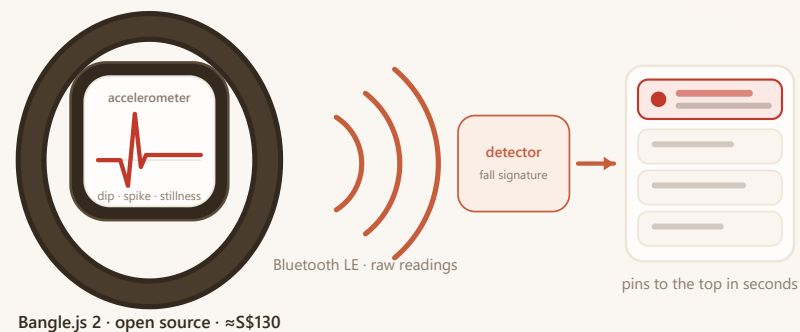
The path is the same in the demo and in deployment. Only the source of the signal changes.

1. The bangle streams accelerometer readings to the scoring service.
2. The detector watches every reading for the ordered fall signature: dip, spike, stillness.
3. A detection pushes an incident event down a live stream the dashboard is always listening to. The resident pins to the top in seconds.

Step 1 is the one link we simulate; there is no wristband in the room.

Each press of the Simulate button feeds a different real SisFall trace straight into step 2; a second button injects a dropped phone the detector ignores. Everything after (detection, push, re-rank) is the production path.

The hardware exists today



Live in the dashboard today: /hardware streams a recorded fall through the production path

DEVICE	REAL PRODUCT, BUYABLE NOW	PRICE
Wearable: raw accelerometer, BLE	Bangle.js 2 , Kionix KX022, open firmware	≈S\$130
Room presence · door events	Aqara P1 motion · contact, 5-yr battery	≈US\$20 · 15
Singapore precedent	GovTech WAAS (iWOW, 2025): 170 blocks · 26,800 seniors	govt tender

BACKUP · FOR Q&A

What a flat **needs**

- Motion sensors in the rooms that carry routine: kitchen, bedroom, bathroom, living room.
- One contact sensor on the front door.
- One optional bangle for the acute track. The routine track works with nothing worn at all.
- An installer fills in a one-page sensor-to-area map. No cameras, no microphones, no rewiring.